

CLINICAL TRIAL PROTOCOL

Code Name: ArthroBalance-X (RA-504)

Study Title: A Phase 3, Multi-center, Randomized Trial for Moderate-to-Severe Rheumatoid Arthritis

Sponsor: Global Pharma Solutions Inc.

Revision: 4.0

Date: January 15, 2026

1. RATIONALE & OBJECTIVES

Rheumatoid Arthritis affects approximately 24 million people globally. Despite advances in biologic therapy, 30-40% of RA patients fail to achieve remission with current TNF- α inhibitors alone. ArthroBalance-X offers a dual mechanism to overcome treatment resistance.

Primary Objective: To demonstrate superior reduction in Disease Activity Score-28 (DAS28) compared to placebo.

Planned Target: Achieve remission (DAS28 <2.6) in 45% of patients by Week 24.

2. STUDY DESIGN

This is a double-blind, randomized, placebo-controlled study across 55 sites globally. The study will enroll 650 participants aged 18-75 with moderate-to-severe active RA (DAS28 >3.2) despite methotrexate therapy. Stratification: By baseline DAS28 score. Study duration: 24 weeks of treatment with 12-week follow-up.

3. SAFETY MONITORING

All participants will undergo baseline and monthly safety assessments including complete blood count and differential. Quarterly assessment of renal function and liver enzymes. Special attention will be paid to serious infections, malignancies, and cardiovascular events.

Adverse Event Tracking: Investigators must report all serious adverse events (SAEs) within 24 hours. The Independent Data Safety Monitoring Committee (IDMC) will perform quarterly unblinded reviews.

Anticipated Effects: Based on Phase 2 data, expected incidence of upper respiratory infections is 12-15%, mild injection site reactions is 8-10%, and herpes zoster reactivation is 2-3%. No tuberculosis reactivation or opportunistic infections were observed in prior phases.

4. ENDPOINTS & STATISTICS

Primary Endpoint: Proportion of participants achieving DAS28 remission (<2.6) at Week 24.

Secondary Endpoints:

- Mean change in DAS28-CRP score
- American College of Rheumatology (ACR) 20/50/70 response rates
- Clinical Disease Activity Index (CDAI) remission rate
- Proportion achieving minimal disease activity (MDA)
- Time to remission
- Change in Health Assessment Questionnaire Disability Index (HAQ-DI)